VOLUME 6

Reference, Glossary, and Index

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-006
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION
The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Original Mazak manual library

Publication	Title	Use
413S038	Operating Manual for VQC-20/40 and 20/50	Primary retained-machine operation reference
M00S075	Operating Manual for CAM M-2	Original conversational control reference
413M033	Maintenance Manual for VQC-20/40 and 20/50	Primary retained mechanical/electrical maintenance
41434WB	Electrical Circuit Diagram for VQC-20/40/50	Primary OEM machine-side wiring source
YM2V39L	PLC Ladder Diagrams	Original sequence and signal meaning
PAREXM210E	Parameter List and Explanation for M-2	M-2 parameter definitions
M00P111	Programming Manual for CAM M-2 EIA/ISO	Original M-2 G-code reference
M00P135	Programming Manual for M-2E	Reference only; machine is M-2
M00P018	Programming Manual for Mazatrol CAM 3D	Option reference; fitment unverified
M00T002	Tooling Manual for Mazak Machining Center	Toolholder and pull-stud reference
413LE02A000	Parts List for VQC 20/40 and 20/50	Mechanical parts sourcing

The 41434WB drawing title-block history includes M-1-era sheets although serial 060231 carries an M-2 control. Use the drawings for retained machine wiring and verify the installed NC/control revision separately.

2. Key project files

File	Authority / purpose
mesa/current_pin_authority.csv	Physical channel authority and evidence state
docs/architecture_decision.md	Selected 7i80HDT/7i44/7i49/two-7i84U architecture
linuxcnc/*.hal and *.ini	Active logic skeleton and configuration placeholders
docs/pre_power_deliverables.md	D1-D16 evidence and hold-point charter
docs/estop_safety_chain.md	Hardware safety requirements and 18-case fault matrix
docs/dc_bus_stop_fault.md	Shared-bus survey/discharge/regen/fault plan
docs/resolver_commissioning.md	Per-axis winding, phasing, scale, and noise proof
docs/servo_commissioning.md	Velocity-mode calibration and FF1-first tuning
docs/hm2_eth_nic_validation.md	Debian 13/NIC/latency/packet/watchdog qualification
docs/frsx_orient_model.md	Discrete-orient model and unresolved FR-SX identity
docs/first_move_plan.md	Signed one-axis first-motion procedure
scripts/validate_authority.py	CSV/HAL/capacity/legend/TB1 regression checks
scripts/validate_control_logic.py	Static HAL/INI safety invariants
scripts/build_manual_set.py	Reproducible Rev B PDF generator

3. Primary external references

Source	Use
Mesa 7i80HDT product page	Host capability: 100BaseT, 72 I/O, three 50-pin connectors, 5 V tolerant
Mesa 7i44 manual	Eight RS-422 channels and physical serial pinout
Mesa 7i49 manual	Six resolver channels, six bipolar analog outputs, cabling, excitation, W2
Mesa 7i84U manual	32 DI/16 DO, TB1/TB2/TB3, 5-28 VDC specification, watchdog/update facts
LinuxCNC hostmot2(9)	Resolver/PWM/sserial/watchdog module semantics
LinuxCNC hm2_eth(9)	Dedicated NIC, packet error, timeout, firewall, coalescing guidance
LinuxCNC pid(9)	FF terms, output engineering units, clamps and windup
Mitsubishi TRA/MELDAS drive manuals	Shared bus, READY, regen, discharge, amplifier setup
Exact Tamagawa suffix datasheets	Required per axis; not yet captured
Exact installed FR-SX manual	Required; model/terminal layout not yet captured

4. Glossary

Term	Definition
AOUT	7i49 bipolar analog output used as a velocity command
Authority	The controlled pin-to-signal record in current_pin_authority.csv
COMMISSIONING_PENDING	Defined design row without completed physical acceptance
D1-D16	Pre-power evidence deliverables and signed hold-point prerequisites
FF1	PID velocity feed-forward term after engineering-unit output scaling
FR-SX	Retained Mitsubishi spindle drive family; exact installed model remains unverified
HAL	LinuxCNC Hardware Abstraction Layer
HOLD_CONFLICT	Contradictory field identity/function; connection or actuation prohibited
hm2_eth	LinuxCNC Ethernet driver for HostMot2 boards
IDROM	Firmware module and pin-routing identity data read from the FPGA image
MAR	OEM relay/chain designation associated with machine-ready/E-stop behavior; trace as-built
ORCM1	Discrete spindle orient command, OEM wire Y093
ORA1	Spindle orient-arrival indication, OEM wire X003
P/N bus	Shared DC bus from retained rectifier/capacitor stack
P3	Unused 7i80HDT 50-pin connector; bare 3.3 V GPIO, no field wiring
PROPOSED	Paper design only; no physical verification
RESDRV	7i49 differential resolver excitation pair
RESSIN/RESCOS	7i49 differential resolver return pairs
S-ON	Servo amplifier enable input
sserial	Mesa smart-serial link; two 7i84Us on 7i44 channels 0/1
TRA	Retained Mitsubishi analog velocity-loop servo amplifier family
TS2014N	Tamagawa shaft-resolver family; exact suffix required per axis
VFIELDA	7i84U field supply for TB3 I/O; TB1 pins 3/4 both positive
VFIELDB	7i84U field supply for TB2 I/O; TB1 pins 1/2 both positive
VIN	7i84U logic supply on TB1 pin 5; W1 can link it to VFIELDB
W1	7i84U jumper selecting linked or separate VIN feed
W2	7i49 half-drive option affecting channels 3/4/5 only
Watchdog	Hardware/remote timeout that must fail outputs off and latch recovery

5. Superseded claim quarantine

Retracted / unsafe claim	Current disposition
7i97T is the primary controller	Retracted; 7i80HDT host with 7i49/7i44 daughters
7i80HD-16 is interchangeable	Contradicted; never flash HD/HD-16 images on HDT
7i37TA is on P3	Retracted; P3 is empty
Probe uses P3 gpio.042	Retracted; 7i84U-B TB3 IN15
Axis feedback is quadrature encoder	Contradicted; resolver through 7i49
AOUT4 commands orient angle	Contradicted; ORCM1 is discrete and AOUT4 spare
W2 can correct X/Y/Z amplitude	Contradicted; W2 affects channels 3/4/5 only
5 kHz is within a published TS2014N tolerance	Unverifiable; exact suffix and scoped measurement required
RESOLVER_SCALE=1.0 is commissioned	Unsafe placeholder; measure signed machine units/electrical rev
OUTPUT_SCALE=MAX_VELOCITY is automatically correct	Only true if measured drive gain makes 10 V equal that velocity
Use fixed P gain 10-30	Rejected; no universal gain; select from measured response
Use 250 ms Z off-delay	Rejected without brake/relay/drive measurements
Probe <=15 IPM is proven safe	Rejected without end-to-end latency, stopping, and repeatability measurements
Disable all NIC RX/TX checksum offloads	Not universal; A/B test supported settings on the installed NIC
7i84U pins 2/4 are returns	Dangerous error; both are positive field-supply duplicates

6. Alphabetical index